Matrice tasti 7 x 8

46 tasti Choc + 9 pulsanti encoder
prese hot-swap Kailh CPG135001S30
un diodo 1N4148W per tasto (anodo verso
la colonna, catodo verso la riga)
scansione: riga bassa, colonne con pull-up

ROW0-6 / COL0-7

U1 Raspberry Pi Pico 2

pin	GPIO	rete
1	GP0	ROW0
2	GP1	ROW1
4	GP2	ROW2
5	GP3	ROW3
6	GP4	ROW4
7	GP5	ROW5
9	GP6	ROW6
10	GP7	COL0
11	GP8	COL1
12	GP9	COL2
14	GP10	COL3
15	GP11	COL4
16	GP12	COL5
17	GP13	COL6
19	GP14	COL7
20	GP15	LED_3V3
21	GP16	SR_LOAD
22	GP17	SR_CLK
24	GP18	SR_DATA
31	GP26	ADC0 (volume)
33	AGND	AGND
35	VREF	ADC_VREF
36	3V3	+3V3
40	VBUS	VBUS
3,8,13,18,23,28,38		GND

LED_3V3 (GP15)

LED (46 x SK6812MINI-E)

GP15 -> U5 74AHCT1G125 (3,3 -> 5 V)
-> R2 330R -> DIN del LED 1
catena a serpentina: pagine/funzioni,
poi voci/trasporto all'indietro, poi TRIG
100nF per LED, firmware limita al 30 %

Encoder (9 x Alps EC11E)

A, B -> pull-up 10k a +3V3 -> 74HC165
C e linguette -> GND
pulsante S1/S2 -> nella matrice
U2: ENC1-4   U3: ENC5-8   U4: DATA
LOAD/CLK in comune, U4->U3->U2->GP18
CLK INH a GND, ingressi liberi a GND

SR_LOAD/CLK/DATA

VBUS / +3V3

Alimentazione

USB del Pico (cavo dal Raspberry Pi 5)
VBUS -> F1 PTC 1,5 A -> +5V
+5V: LED, U5, J1 (display)
C70 220uF + 100nF locali
+3V3 dal Pico: 74HC165, pull-up
AGND unita a GND con R3 0R

ADC0 (GP26)

Volume (RV1 Alps RK09K 10k)

pin 1 -> AGND
pin 3 -> ADC_VREF
cursore -> R1 1k -> GP26
C60 100nF verso AGND (filtro)

Connettori

USB micro del Pico: al Raspberry Pi 5
J1 2 pin 2,54 mm: 1 = +5V, 2 = GND
   cavo di alimentazione del display
H1-H10: fori M3 sui prigionieri del
   pannello, collegati a GND

DRUMMATIC

Schema a blocchi e collegamenti del Pico 2

Circuito di controllo · rev. A · 2026-10 · KiCad 7

	COL0 GP7	COL1 GP8	COL2 GP9	COL3 GP10	COL4 GP11	COL5 GP12	COL6 GP13	COL7 GP14
ROW0 GP0	VOICE nota 0	SMPL nota 1	TONE nota 2	MOD nota 3	SEQ nota 4	LANE nota 5	MIX nota 6	PERF nota 7
ROW1 GP1	FUNC nota 8	MAP nota 9	LOCK nota 10	DADO nota 11	COMMIT nota 12	KIT nota 13	COPY nota 14	DATA (premuto) nota 72
ROW2 GP2	VOICE1 nota 15	VOICE2 nota 16	VOICE3 nota 17	VOICE4 nota 18	VOICE5 nota 19	VOICE6 nota 20	VOICE7 nota 21	VOICE8 nota 22
ROW3 GP3	REC nota 23	PLAY nota 24	STOP nota 25	LEFT nota 26	UP nota 27	DOWN nota 28	RIGHT nota 29	—
ROW4 GP4	TRIG1 nota 30	TRIG2 nota 31	TRIG3 nota 32	TRIG4 nota 33	TRIG5 nota 34	TRIG6 nota 35	TRIG7 nota 36	TRIG8 nota 37
ROW5 GP5	TRIG9 nota 38	TRIG10 nota 39	TRIG11 nota 40	TRIG12 nota 41	TRIG13 nota 42	TRIG14 nota 43	TRIG15 nota 44	TRIG16 nota 45
ROW6 GP6	ENC1 (premuto) nota 64	ENC2 (premuto) nota 65	ENC3 (premuto) nota 66	ENC4 (premuto) nota 67	ENC5 (premuto) nota 68	ENC6 (premuto) nota 69	ENC7 (premuto) nota 70	ENC8 (premuto) nota 71

Ogni incrocio: colonna -> contatto della presa hot-swap -> diodo 1N4148W (anodo) -> catodo -> riga.

Messaggi USB-MIDI del firmware, canale 16: tasto premuto = Note On (velocity 127), rilasciato = Note Off.

Encoder 1-8 = CC 16-23, DATA = CC 24 (relativi: 1-63 destra, 65-127 sinistra); volume = CC 7.

LED: Note On n con velocity = colore (0 spento, 1-15 tavolozza, +16 metà luce); CC 119 luminosità; CC 121 tutto spento.

U2 74HC165

A (11) ENC1_A
B (12) ENC1_B
C (13) ENC2_A
D (14) ENC2_B
E (3)   ENC3_A
F (4)   ENC3_B
G (5)   ENC4_A
H (6)   ENC4_B
SER(10) SR_S1
QH (9) SR_DATA (GP18)
SH/LD(1) SR_LOAD CLK(2) SR_CLK
CLK INH(15) GND VCC(16) +3V3 GND(8)

U3 74HC165

A (11) ENC5_A
B (12) ENC5_B
C (13) ENC6_A
D (14) ENC6_B
E (3)   ENC7_A
F (4)   ENC7_B
G (5)   ENC8_A
H (6)   ENC8_B
SER(10) SR_S2
QH (9) SR_S1 -> SER di U2
SH/LD(1) SR_LOAD CLK(2) SR_CLK
CLK INH(15) GND VCC(16) +3V3 GND(8)

U4 74HC165

A (11) DATA_A
B (12) DATA_B
C (13) GND
D (14) GND
E (3)   GND
F (4)   GND
G (5)   GND
H (6)   GND
SER(10) GND
QH (9) SR_S2 -> SER di U3
SH/LD(1) SR_LOAD CLK(2) SR_CLK
CLK INH(15) GND VCC(16) +3V3 GND(8)

Catena LED (ordine DIN -> DOUT)

1:VOICE, 2:SMPL, 3:TONE, 4:MOD, 5:SEQ, 6:LANE, 7:MIX, 8:PERF,
9:FUNC, 10:MAP, 11:LOCK, 12:DADO, 13:COMMIT, 14:KIT, 15:COPY,
16:RIGHT, 17:DOWN, 18:UP, 19:LEFT, 20:STOP, 21:PLAY, 22:REC,
23:VOICE8, 24:VOICE7, 25:VOICE6, 26:VOICE5, 27:VOICE4,
28:VOICE3, 29:VOICE2, 30:VOICE1, 31:TRIG1, 32:TRIG2, 33:TRIG3,
34:TRIG4, 35:TRIG5, 36:TRIG6, 37:TRIG7, 38:TRIG8, 39:TRIG9,
40:TRIG10, 41:TRIG11, 42:TRIG12, 43:TRIG13, 44:TRIG14,
45:TRIG15, 46:TRIG16

Distinta componenti

55	x	1N4148W	1N4148W-7-F (Diodes Inc.)	D_SOD-123
51	x	100nF	GRM188R71H104KA93D	C_0603_1608Metric
46	x	Choc v1	Kailh CPG135001S30 (presa hot-swap)	Choc_v1_Hotswap
46	x	SK6812MINI-E	SK6812MINI-E (Opsco)	SK6812MINI-E_Reverse
18	x	10k	RC0603FR-0710KL	R_0603_1608Metric
10	x	M3		MountingHole_3.2mm_M3_Pad
9	x	EC11E	Alps EC11E15244G1 (o EC11E15244B2)	RotaryEncoder_Alps_EC11E-Switch_Vertical_H20mm
3	x	74HC165	SN74HC165DR (TI)	SOIC-16_3.9x9.9mm_P1.27mm
1	x	10k lin VOLUME	Alps RK09K1130A5R (10k B)	Potentiometer_Alps_RK09K_Single_Vertical
1	x	1k	RC0603FR-071KL	R_0603_1608Metric
1	x	Raspberry Pi Pico 2	Raspberry Pi Pico 2 (SC1631)	RPi_Pico_SMD_TH
1	x	PTC 1.5A	Littelfuse 1206L150THYR	Fuse_1206_3216Metric
1	x	220uF 10V	Panasonic EEE-FK1A221P	CP_Elec_6.3x7.7
1	x	74AHCT1G125	SN74AHCT1G125DBVR (TI)	SOT-23-5
1	x	330R	RC0603FR-07330RL	R_0603_1608Metric
1	x	DISPLAY 5V	Header 2 pin 2,54 mm	PinHeader_1x02_P2.54mm_Vertical
1	x	0R	RC0603JR-070RL	R_0603_1608Metric